

Hands-on 1: Angular Project Setup and Component Creation

Objective:

To create an Angular application, understand the project structure, create reusable components, and build a basic application layout.

1. Angular Project Setup

First, we created a new Angular project using Angular CLI.

Command used:

```
ng new employee-app
```

This command creates a new Angular application with all the required files and dependencies.

Then we entered the project folder:

```
cd employee-app
```

To run the application:

```
ng serve
```

ng serve starts the Angular development server and runs the application locally.

The application can be accessed using:

```
http://localhost:4200/
```

2. Understanding Angular Project Structure

Important folders/files:

```
src/  
├─ app/  
│   ├── components/  
│   ├── pages/  
│   ├── app.ts  
│   ├── app.html  
│   └─ app.css
```

src/

Contains the main source code of the application.

app/

Contains Angular application components and logic.

app.ts

Contains the root component TypeScript logic.

app.html

Contains the root component HTML template.

app.css

Contains styling for the root component.

3. Creating Components

Angular components are reusable building blocks of an application.

A component contains:

- TypeScript file → Logic
- HTML file → UI structure
- CSS file → Styling

Created Header component:

Command:

```
ng generate component components/header
```

Created Home component:

```
ng generate component pages/home
```

4. Using Components in Application

Each component has a selector.

Example:

```
selector: 'app-header'
```

The component can be displayed using:

```
<app-header></app-header>
```

The root component imports the required components:

```
imports: [  
  Header,  
  Home  
]
```

This allows Angular to recognize and use these components.

5. Building Basic Application Layout

Created:

Header Component

Contains navigation:

- Student Course Portal title
- Home
- Courses
- Profile links

Home Component

Displays:

- Welcome message
- Basic course information

The final application structure:

